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P R O J E C T R E P O R T

Diabetes Onset Prediction

Using Machine Learning

An End-to-End Predictive Analytics / KDD Study
on the Pima Indians Diabetes Dataset

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Project: End-to-End Predictive Analytics Model (KDD process)
Dataset: Pima Indians Diabetes Dataset (768 records, 8 features)
Submitted: October 2026

Disclaimer: *This is an academic project. The model described here is not a medical diagnostic system and has not been clinically validated.*

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Abstract

This project develops an end-to-end predictive model that estimates whether a patient is likely to have diabetes from eight routine diagnostic measurements in the Pima Indians Diabetes Dataset. Following the Knowledge Discovery in Databases (KDD) process, physiologically impossible zero values were treated as missing data, the data were split with stratification before any fitting, and median imputation, standardisation and SMOTE oversampling were learned from the training data only, inside a leak-free pipeline. Five classifiers were compared: Logistic Regression, K-Nearest Neighbors, Decision Tree, Random Forest and XGBoost. They were evaluated with 5-fold cross-validation and on a held-out test set of 154 patients. Using a recall-aware selection rule, **Random Forest** was chosen as the final model, with test accuracy 0.747, precision 0.615, recall 0.741, F1 0.672 and ROC-AUC 0.809 (cross-validated ROC-AUC 0.829). Feature importance and SHAP analysis consistently identified **plasma glucose** as the dominant predictor, followed by BMI, age and the diabetes pedigree function.

Chapter 1 — Introduction

1.1 Background

Diabetes mellitus is a chronic metabolic disease characterised by raised blood glucose, caused by insufficient insulin production, reduced sensitivity to insulin, or both. The World Health Organization describes diabetes as a major cause of blindness, kidney failure, heart attacks, stroke and lower-limb amputation [1]. Type 2 diabetes, the most common form, often develops gradually and may remain undiagnosed for years, so many people are identified only after complications have begun.

Health systems already collect simple measurements such as plasma glucose, body-mass index (BMI), blood pressure and age. Predictive analytics uses historical records of such measurements with known outcomes to learn patterns that can estimate risk for new individuals. In a screening setting, a model of this kind could help prioritise who should undergo confirmatory clinical testing.

1.2 Problem Statement

Given a vector of diagnostic measurements **X** = (Pregnancies, Glucose, BloodPressure, SkinThickness, Insulin, BMI, DiabetesPedigreeFunction, Age) for a patient, predict the binary outcome $\mathbf{y \in \{0, 1\}}$, where 1 indicates diabetes, and estimate the probability $P(y = 1 \mid X)$. This is a **supervised binary classification** problem on imbalanced, partially incomplete tabular data.

1.3 Motivation

- **Early identification.** Earlier detection of high-risk individuals allows earlier confirmatory testing and lifestyle or medical intervention.
- **Low-cost inputs.** The features are routine measurements, so a screening-support model would not require expensive tests.
- **Academic value.** The dataset contains realistic data-quality problems: hidden missing values encoded as zeros, class imbalance and skewed features. This makes it well suited to demonstrating every stage of the KDD process.

1.4 Objectives

1. Understand and clean the dataset, including identifying invalid zero values.
2. Explore relationships between the measurements and the outcome using visual EDA.
3. Build a leak-free preprocessing pipeline (imputation, scaling, SMOTE) fitted on training data only.
4. Train and compare Logistic Regression, KNN, Decision Tree, Random Forest and XGBoost.
5. Evaluate the models with Accuracy, Precision, Recall, F1, ROC-AUC, confusion matrices and ROC curves.

6. Select a final model with a justified, screening-oriented criterion rather than accuracy alone.
7. Interpret the model with feature importance and SHAP, and discuss limitations.

1.5 Scope

The project covers the full modelling cycle on a single public dataset, from problem definition to interpretation. It produces an academic screening model and a demonstration prediction function. It does **not** include clinical validation, external validation on other populations, deployment, or exhaustive hyperparameter optimisation.

1.6 Dataset Overview

The Pima Indians Diabetes Dataset was originally collected by the US National Institute of Diabetes and Digestive and Kidney Diseases and first used in machine-learning research by Smith et al. [2]. It was historically distributed through the UCI Machine Learning Repository and is now widely available via Kaggle [3], [4]. The copy used here comes from the public repository of J. Brownlee [5]. All 768 subjects are females of Pima Indian heritage aged at least 21.

Feature	Description	Unit
Pregnancies	Number of times pregnant	count
Glucose	Plasma glucose concentration, 2 h in an oral glucose tolerance test	mg/dL
BloodPressure	Diastolic blood pressure	mm Hg
SkinThickness	Triceps skin-fold thickness	mm
Insulin	2-hour serum insulin	µU/mL
BMI	Body-mass index	kg/m²
DiabetesPedigreeFunction	Score summarising diabetes history in relatives	–
Age	Age	years
Outcome	**0 = no diabetes, 1 = diabetes**	class

The dataset has 768 rows, 9 numeric columns, no NaN values, no duplicate rows, and a class distribution of **500 (65.1 %) non-diabetic and 268 (34.9 %) diabetic** patients.

Chapter 2 — Literature Review

The Pima dataset is one of the most widely used benchmarks for diabetes prediction. This chapter summarises representative published studies. The figures quoted below are **the findings reported by those authors**, obtained under their own preprocessing, splits and validation schemes. They are **not** results of this project and are not directly comparable to them.

2.1 Smith et al. (1988): ADAP neural network [2]

- **Problem:** Forecast the onset of diabetes mellitus within five years in a high-risk population.
- **Dataset:** The original Pima Indian data from the NIDDK (the source of the dataset used here). As described in the dataset documentation, 576 cases were used for training and 192 for testing.
- **Method:** ADAP, an early adaptive-learning neural-network-like algorithm, compared with logistic regression and a linear perceptron.
- **Main finding:** Sensitivity and specificity crossed at about **0.76**.
- **Relevance:** This study introduced the dataset and established that the measurements carry predictive signal. It also evaluated performance with sensitivity, specificity and ROC analysis rather than accuracy alone, the same reasoning applied in this project.

2.2 Maniruzzaman et al. (2017): Gaussian process classification [6]

- **Problem:** Compare machine-learning paradigms for classifying diabetes data.
- **Dataset:** Pima Indian diabetes dataset (768 subjects: 268 diabetic, 500 controls).
- **Method:** Gaussian-process (GP) classification with linear, polynomial and radial-basis kernels, compared with linear discriminant analysis, quadratic discriminant analysis and naïve Bayes. Evaluation used accuracy, sensitivity, specificity, PPV, NPV and ROC curves.
- **Main finding:** The GP-based model performed best. Its reported accuracy, sensitivity and specificity were about 82 %, 92 % and 63 %.
- **Relevance:** This study shows the trade-off between sensitivity and specificity on this dataset and the value of reporting several metrics, both central themes of this project.

2.3 Sisodia and Sisodia (2018): Comparison of classification algorithms [7]

- **Problem:** Predict diabetes at an early stage with standard classification algorithms.
- **Dataset:** Pima Indians Diabetes Dataset.
- **Method:** Decision Tree, Support Vector Machine and Naïve Bayes, evaluated with precision, recall, F-measure and accuracy.
- **Main finding:** Naïve Bayes achieved the highest reported accuracy, about 76.3 %.

- **Relevance:** This is a baseline comparison of simple classifiers on the same data. It motivates the comparison of several model families in this project and shows that a single decision tree is typically not the strongest model.

2.4 Zou et al. (2018): Decision trees, random forests and neural networks [8]

- **Problem:** Predict diabetes mellitus from routine physical-examination data.
- **Dataset:** Hospital physical-examination records from Luzhou, China (14 attributes). The Pima Indians dataset was used as a second dataset, reduced to 392 complete records after removing missing values.
- **Method:** Decision tree, random forest and neural network, with dimensionality reduction by PCA and mRMR feature selection, evaluated with five-fold cross-validation.
- **Main finding:** Random forest gave the highest accuracy: 0.8084 on the Luzhou data with all attributes, and 0.7604 on the Pima data (0.7721 with mRMR). Glucose alone gave reasonable accuracy on the Luzhou data, but the authors concluded that glucose alone was not sufficient for the best result.
- **Relevance:** This supports including Random Forest as a strong candidate and shows that, although glucose is the key predictor, combining it with other measurements improves prediction, consistent with this project's findings. It also handled missing Pima values by deleting rows, whereas this project uses imputation to keep all 768 records.

2.5 Hasan et al. (2020): Ensembling of machine-learning classifiers [9]

- **Problem:** Build a robust diabetes-prediction framework.
- **Dataset:** Pima Indian Diabetes Dataset.
- **Method:** Outlier rejection, missing-value filling, standardisation and feature selection, followed by k-NN, decision trees, random forest, AdaBoost, naïve Bayes, XGBoost and an MLP. Models were combined with AUC-based weights and evaluated with K-fold cross-validation.
- **Main finding:** The proposed AUC-weighted ensemble was the best-performing classifier, with a reported AUC of 0.950, sensitivity 0.789 and specificity 0.934.
- **Relevance:** This study demonstrates the importance of preprocessing (especially handling missing values) and of ensemble methods such as XGBoost, both used in this project. Its much higher AUC comes from a more elaborate pipeline (outlier removal, feature selection, ensembling) and a different validation protocol, so it is not directly comparable.

2.6 Joshi and Dhakal (2021): Logistic regression and decision trees [10]

- **Problem:** Predict type 2 diabetes in Pima Indian women and identify its main predictors.
- **Dataset:** Pima Indians Diabetes Dataset.

- **Method:** Logistic regression for predictor selection, a decision tree, and cross-validation.
- **Main finding:** The preferred specification reached a prediction accuracy of 78.26 % (cross-validation error 21.74 %). Glucose, pregnancies, BMI, diabetes pedigree function and age were identified as significant predictors.
- **Relevance:** This study supports using Logistic Regression as an interpretable baseline, and its predictor list closely matches the most important features found in this project.

2.7 Kavakiotis et al. (2017): Review of machine learning in diabetes research [11]

- **Problem:** Survey how machine learning and data mining are applied in diabetes research.
- **Dataset:** A systematic review of many studies, not a single dataset.
- **Method:** Literature review covering prediction and diagnosis, complications, genetics, and healthcare management.
- **Main finding:** Prediction and diagnosis is the most common application, and supervised methods (especially support vector machines) dominate. Random forests and other ensembles are also widely used.
- **Relevance:** This review places the project within the broader research landscape and justifies the choice of supervised classifiers.

2.8 Methodological foundations

- **KDD process.** Fayyad, Piatetsky-Shapiro and Smyth [12] defined KDD as the overall process of discovering useful knowledge from data: selection, preprocessing, transformation, data mining and interpretation. This project is structured around those stages.
- **SMOTE.** Chawla et al. [13] proposed generating synthetic minority-class samples by interpolating between neighbouring minority examples. This improves classifier sensitivity to the minority class compared with plain oversampling.
- **Random Forest.** Breiman [14] introduced random forests: ensembles of decision trees trained on bootstrap samples with random feature selection at each split. They reduce variance compared with a single tree.
- **XGBoost.** Chen and Guestrin [15] described a scalable, regularised gradient-tree-boosting system that is widely used for tabular data.
- **SHAP.** Lundberg and Lee [16] unified several explanation methods under Shapley values from cooperative game theory. This gives additive per-feature contributions to each prediction.
- **Software.** Scikit-learn [17] and imbalanced-learn [18] provide the implementations of pipelines, models and SMOTE used here.

2.9 Research gap addressed by this project

Published accuracies on the Pima dataset vary widely (roughly 76 % to above 90 %), partly because studies differ in how they treat the invalid zeros, whether they apply oversampling or scaling before splitting the data, and how they validate. This project does not aim to beat published numbers. Instead, it aims to produce a **transparent, leak-free and reproducible** baseline, in which every learned transformation (imputation, scaling, SMOTE) is fitted only on training data, and model selection explicitly considers recall.

Chapter 3 — Workflow / Predictive Modelling Process

The project follows the KDD process (Figure 3.1). Each stage corresponds to a section of the notebook.

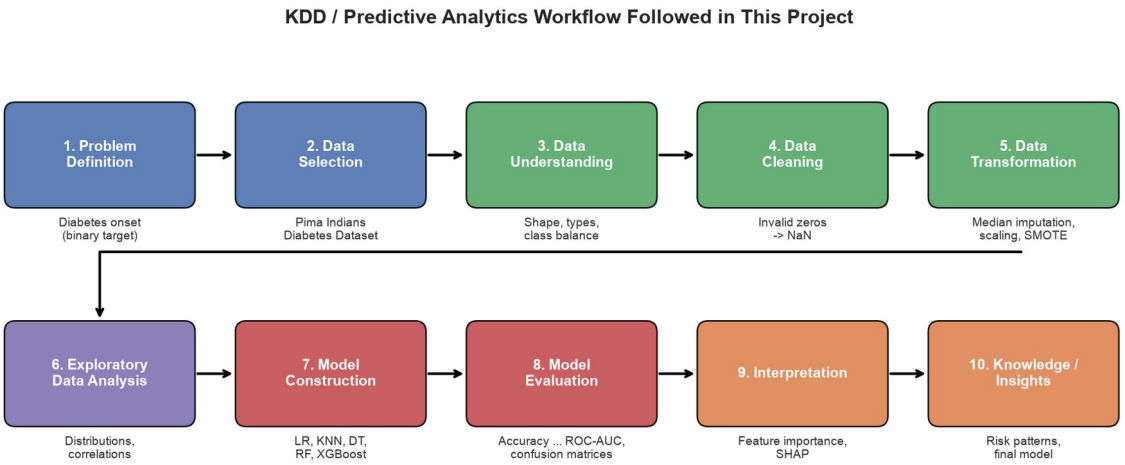


Figure 3.1: KDD / predictive analytics workflow followed in this project.

```
Problem Definition → Data Selection → Data Understanding → Data Cleaning → Data
                                    Transformation
    → Exploratory Data Analysis → Model Construction → Model Evaluation →
                    Interpretation → Knowledge / Insights
```

3.1 Problem Definition

The task was framed as binary classification (Outcome 0/1) with a probability output. Because the intended use is *screening support*, missing a diabetic patient (false negative) was treated as more costly than a false alarm. Recall was therefore given special weight alongside the threshold-independent ROC-AUC.

3.2 Data Collection

The Pima Indians Diabetes Dataset was downloaded from the public repository [5]. The original file has no header, so the nine column names were added. The 768 observations were not altered, and all cleaning is performed in code. The notebook resolves the data path automatically and can re-download the file if it is missing.

3.3 Data Understanding

`head()`, `tail()`, `info()`, `describe()`, `dtypes`, missing-value, duplicate and unique-value checks were performed. All features are numeric and there are no NaNs or duplicate rows. However, `describe()` revealed **minimum values of 0** in Glucose, BloodPressure, SkinThickness, Insulin and BMI, which is physiologically impossible. The target is imbalanced (65.1 % / 34.9 %).

3.4 Data Cleaning

Zero values in the five affected columns were replaced with NaN. Zeros in Pregnancies (a valid count) and Outcome (the negative class) were kept.

Feature	Invalid zeros	% of rows
Glucose	5	0.65
BloodPressure	35	4.56
SkinThickness	227	29.56
Insulin	374	48.70
BMI	11	1.43

Replacing 0 with NaN is a fixed rule that learns nothing from the data, so applying it before splitting cannot leak information. Rows were not dropped, because removing every incomplete row would discard roughly half the dataset.

3.5 Exploratory Data Analysis

EDA included the class count plot, histograms of all features, by-outcome density plots for Glucose, BMI and Age, boxplots of all features by outcome, a correlation heatmap and a pairplot of the four features most correlated with the outcome. Every plot has a title, axis labels and a legend where relevant. Findings are reported in Chapter 4.

3.6 Data Preprocessing

1. **Stratified split first.** `train_test_split(X, y, test_size=0.2, stratify=y, random_state=42)` produced 614 training rows (400 / 214) and 154 test rows (100 / 54). Both sets keep the original class ratio of about 35 % positives.
2. **Median imputation**, with medians learned from the training set (e.g. Insulin 125.0, SkinThickness 29.0, Glucose 117.0). The median was chosen because Insulin and SkinThickness are strongly right-skewed.
3. **Standardisation** (`StandardScaler`), with means and standard deviations learned from the training set.

The test set never influences any learned parameter. Otherwise, information about "future" patients would leak into training and inflate the evaluation scores.

3.7 Class Imbalance

SMOTE [13] was used to oversample the diabetic class **in the training data only**, from 400 / 214 to 400 / 400. The test set keeps its real 100 / 54 distribution. SMOTE was placed *inside* an `imblearn` pipeline after imputation and scaling, so during cross-validation it is re-applied to each training fold and never touches the validation fold.

`class_weight="balanced"` was compared as an alternative, using Logistic Regression and 5-fold CV on the training data:

Strategy	CV Accuracy	CV Recall	CV F1	CV ROC-AUC
No imbalance handling	0.7882	0.5748	0.6545	0.8434
`class_weight="balanced"`	0.7622	0.7010	0.6724	0.8444
SMOTE	0.7720	0.7244	0.6885	0.8427

Without imbalance handling, recall falls to 0.57, meaning that more than 4 in 10 diabetic patients would be missed. Both techniques raise recall by about 13–15 points at a small cost in accuracy, while ROC-AUC (a ranking measure) is essentially unchanged. SMOTE was adopted for all models because it applies uniformly, including to KNN, which has no class-weight option. Class weights were not combined with SMOTE, to avoid correcting the imbalance twice.

3.8 Model Selection

Five algorithms representing different learning paradigms were chosen:

Model	Paradigm	Configuration
Logistic Regression	Linear, probabilistic baseline	`max_iter=1000`
K-Nearest Neighbors	Instance-based (distance)	`n_neighbors=15`
Decision Tree	Rule-based tree	`max_depth=5`, `min_samples_leaf=5`
Random Forest	Bagging ensemble [14]	`n_estimators=300`, `min_samples_leaf=2`
XGBoost	Gradient-boosting ensemble [15]	`n_estimators=200`, `max_depth=3`, `learning_rate=0.05`, `subsample=0.8`, `colsample_bytree=0.8`

The settings are conservative defaults chosen to limit overfitting on a small dataset. No hyperparameters were tuned on the test set.

3.9 Model Training

Every model uses the same pipeline:

```
SimpleImputer(median) → StandardScaler → SMOTE → Classifier
```

Scaling is required for Logistic Regression and KNN. It is harmless for the tree models, because it does not change the possible splits, and it gives SMOTE comparable feature scales for its nearest-neighbour search. Each pipeline was first assessed with **5-fold stratified cross-validation on the training set**, then refitted on all 614 training rows. `random_state=42` was used throughout.

3.10 Evaluation

The fitted pipelines were evaluated once on the 154-patient test set using Accuracy, Precision, Recall, F1, ROC-AUC and Specificity, together with classification reports, confusion matrices and ROC curves. The final model was chosen with a predefined rule:

1. **Screening constraint:** recall ≥ 0.70 in both cross-validation and the test set.

2. **Overall score:** the mean of the five metrics, averaged equally over cross-validation and test. This rewards consistency rather than performance on one split.

3.11 Interpretation

The final model was interpreted with (a) impurity-based feature importance from the three tree models and the absolute standardised coefficients of Logistic Regression, and (b) SHAP values computed with `TreeExplainer` [16] on the test set. A permutation-importance fallback was implemented in case SHAP failed, but it was not needed.

Chapter 4 — Results and Discussion

All values in this chapter come from the executed notebook and the files `results/model_comparison.csv` and `results/metrics.csv`.

4.1 Dataset Statistics

Feature	Mean	Std	Min	Median	Max
Pregnancies	3.85	3.37	0	3.0	17
Glucose	120.89	31.97	0*	117.0	199
BloodPressure	69.11	19.36	0*	72.0	122
SkinThickness	20.54	15.95	0*	23.0	99
Insulin	79.80	115.24	0*	30.5	846
BMI	31.99	7.88	0*	32.0	67.1
DiabetesPedigreeFunction	0.47	0.33	0.078	0.37	2.42
Age	33.24	11.76	21	29.0	81

** Invalid zeros, treated as missing values. Statistics are computed on the raw data, before cleaning.*

4.2 Class Distribution

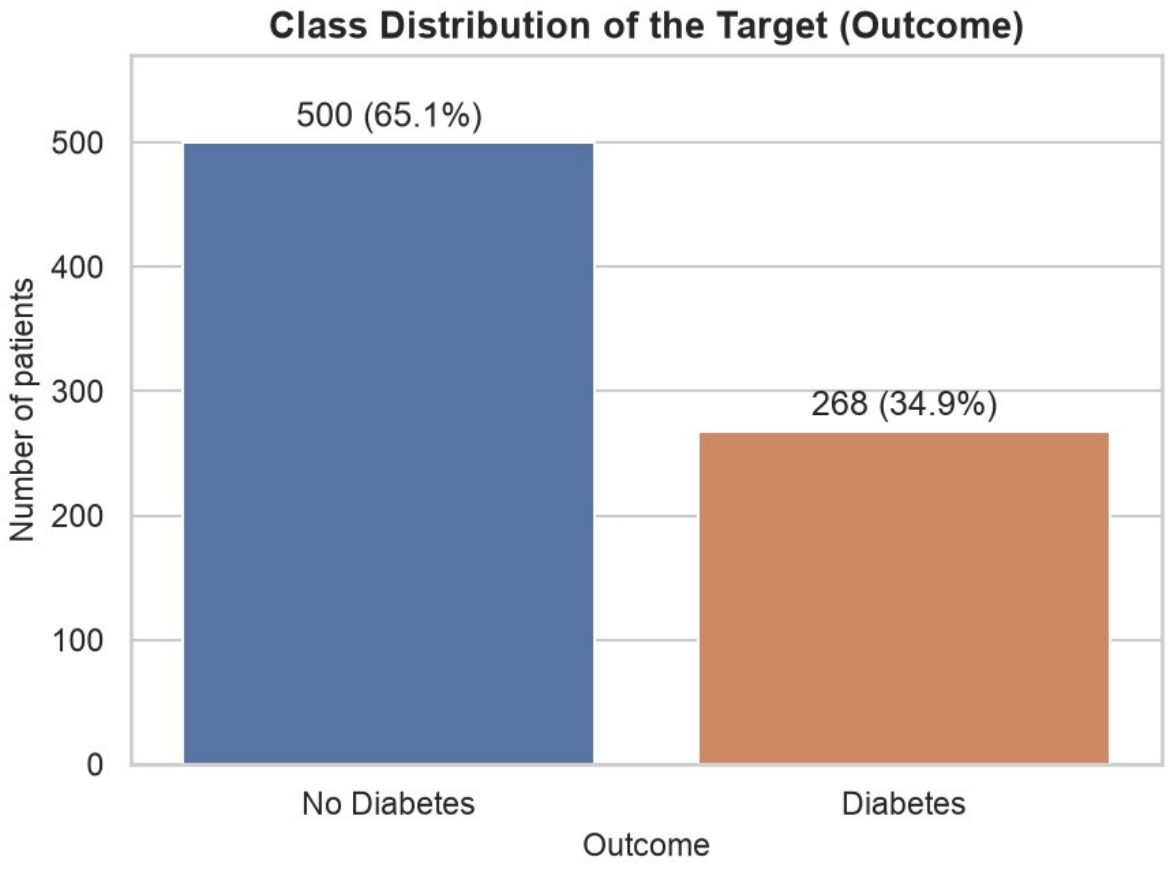


Figure 4.1: 500 non-diabetic (65.1 %) vs. 268 diabetic (34.9 %) patients.

A trivial classifier that always predicts "no diabetes" would score 65 % accuracy while detecting no diabetic patients. This is why accuracy is not used on its own.

4.3 EDA Findings

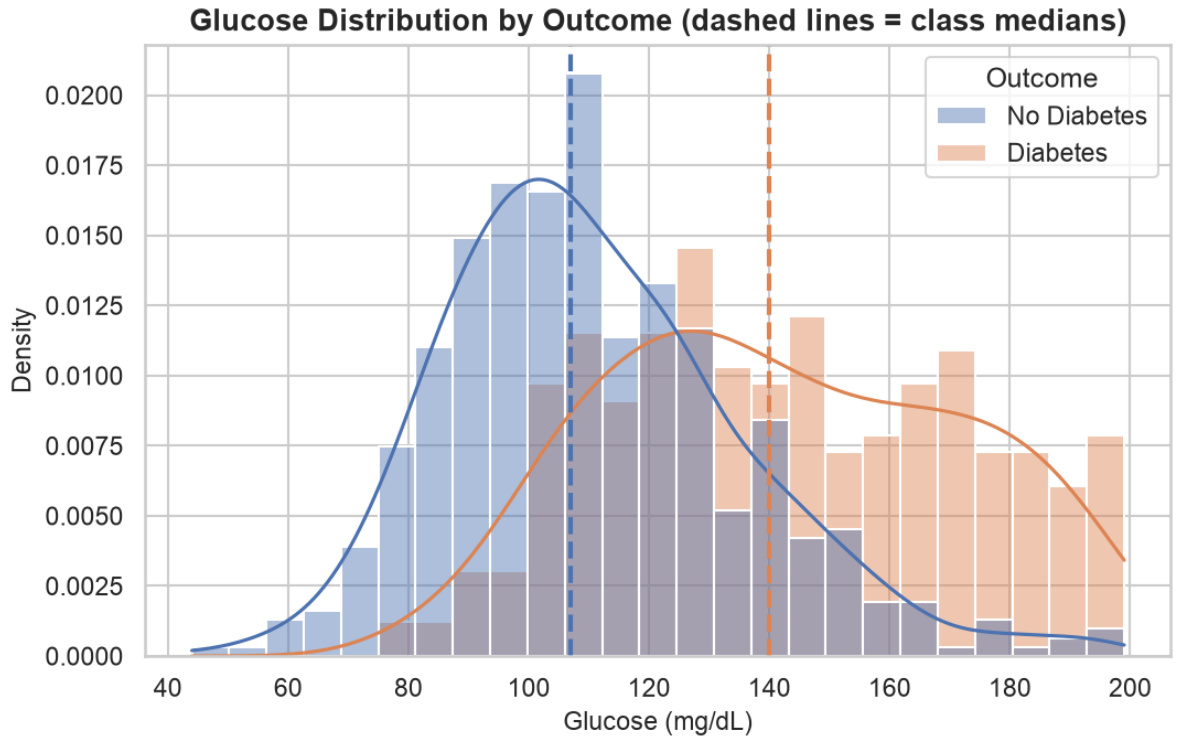


Figure 4.2: Glucose distribution by outcome (dashed lines = class medians).

Feature	Median (No Diabetes)	Median (Diabetes)	Correlation with Outcome
Glucose	107.0	140.0	0.495
BMI	30.1	34.3	0.314
Insulin	102.5	169.5	0.303
SkinThickness	27.0	32.0	0.259
Age	27.0	36.0	0.238
Pregnancies	2.0	4.0	0.222
DiabetesPedigreeFunction	0.34	0.45	0.174
BloodPressure	70.0	74.5	0.171

(Computed after replacing invalid zeros with NaN.)

- **Glucose** separates the classes most clearly (medians of 140 vs. 107 mg/dL; r = 0.495).
- **BMI** and **Age** are moderately higher in diabetic patients. Non-diabetic patients are concentrated in their 20s.
- **Insulin** and **SkinThickness** are higher in the diabetic group but have the most missing data.

- The strongest feature–feature correlations are SkinThickness–BMI (0.65), Glucose–Insulin (0.58) and Pregnancies–Age (0.54). These are moderate and do not justify removing features.
- Every feature shows substantial class overlap, so no single measurement is sufficient for classification.

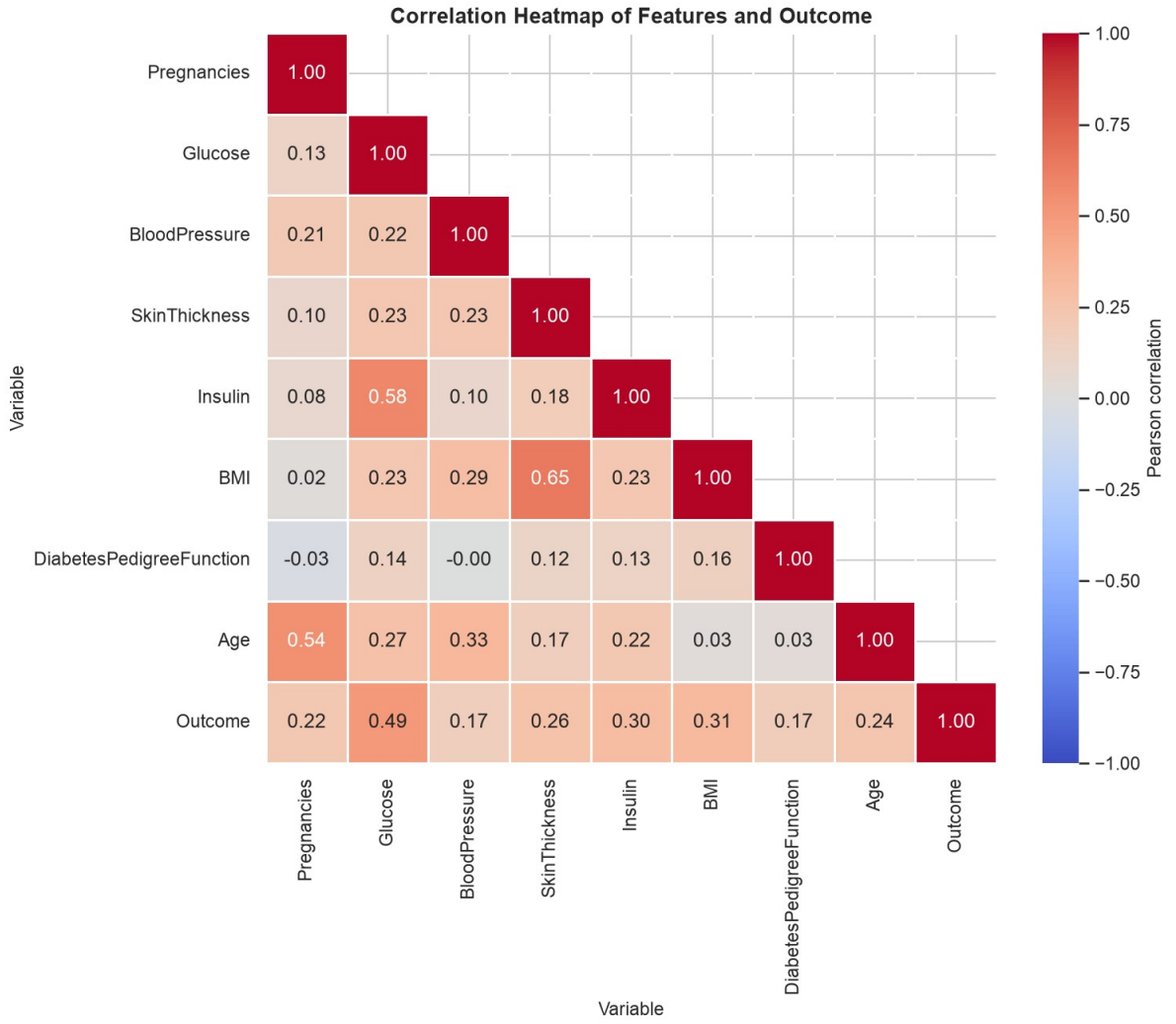


Figure 4.3: Correlation heatmap (pairwise-complete observations).

These relationships are **associations** in this sample, not evidence of causation.

4.4 Model Comparison

Cross-validation on the training set (5 folds, mean ± std):

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	0.772 ± 0.015	0.658 ± 0.026	0.724 ± 0.045	0.689 ± 0.021	0.843 ± 0.020
K-Nearest Neighbors	0.728 ± 0.054	0.588 ± 0.070	0.790 ± 0.020	0.672 ± 0.047	0.820 ± 0.045
Decision Tree	0.697 ± 0.029	0.560 ± 0.047	0.687 ± 0.069	0.612 ± 0.013	0.762 ± 0.013
Random Forest	0.779 ± 0.009	0.672 ± 0.022	0.715 ± 0.028	0.692 ± 0.009	0.829 ± 0.025

Model	Accuracy	Precision	Recall	F1	ROC-AUC
XGBoost	0.751 ± 0.027	0.632 ± 0.041	0.687 ± 0.058	0.657 ± 0.040	0.822 ± 0.030

Hold-out test set (154 patients), sorted by ROC-AUC (`results/model_comparison.csv`):

Model	Accuracy	Precision	Recall	F1	ROC-AUC	Specificity
XGBoost	0.7532	0.6250	0.7407	0.6780	0.8263	0.760
Logistic Regression	0.7078	0.5738	0.6481	0.6087	0.8113	0.740
Random Forest	0.7468	0.6154	0.7407	0.6723	0.8094	0.750
K-Nearest Neighbors	0.6948	0.5455	0.7778	0.6412	0.7974	0.650
Decision Tree	0.7273	0.5833	0.7778	0.6667	0.7616	0.700

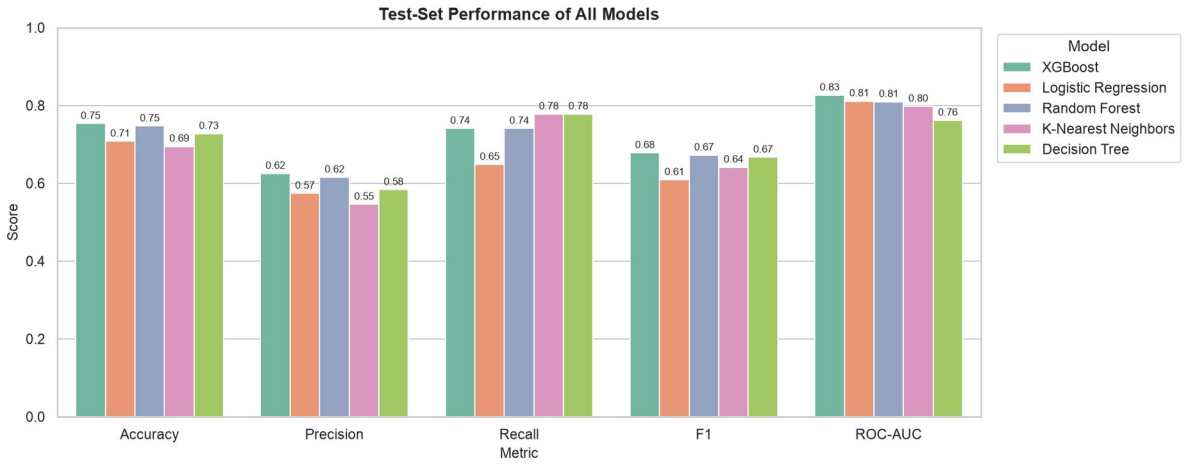


Figure 4.4: Test-set metrics for all five models.

Best model per metric (test set): Accuracy: XGBoost (0.7532). Precision: XGBoost (0.6250). Recall: KNN and Decision Tree (0.7778). F1: XGBoost (0.6780). ROC-AUC: XGBoost (0.8263).

4.5 Confusion Matrices

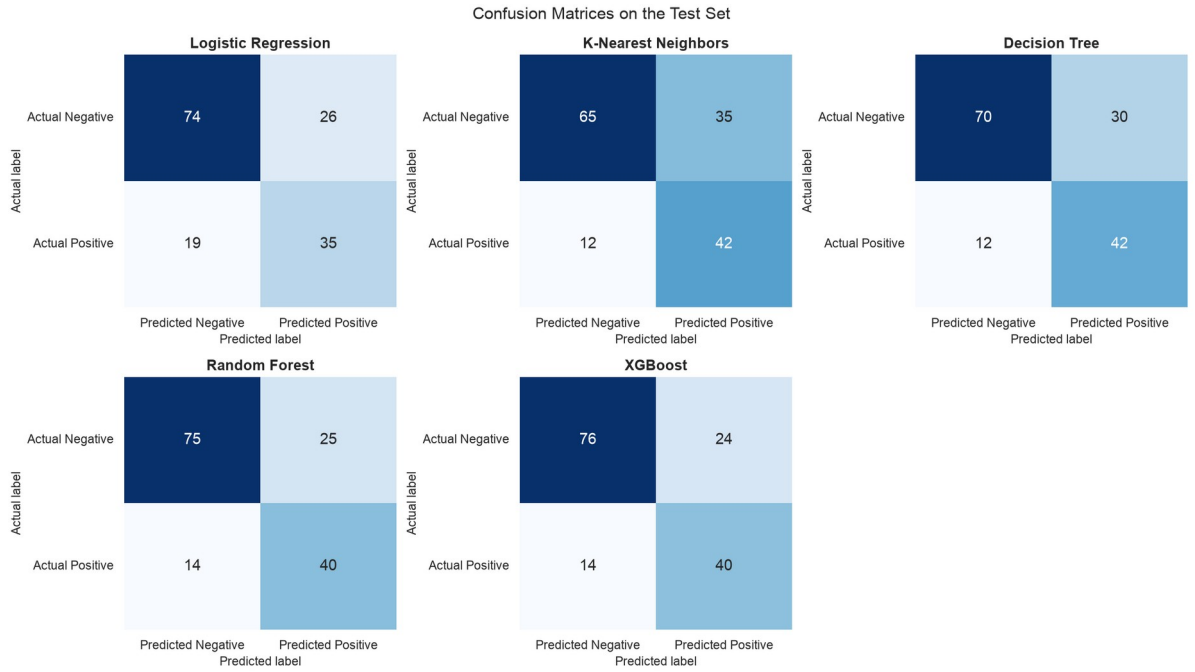


Figure 4.5: Confusion matrices on the test set (54 diabetic, 100 non-diabetic patients).

Model	TP	TN	FP	FN
XGBoost	40	76	24	14
Logistic Regression	35	74	26	19
Random Forest	40	75	25	14
K-Nearest Neighbors	42	65	35	12
Decision Tree	42	70	30	12

KNN and the Decision Tree miss the fewest diabetic patients (12 FN) but raise the most false alarms (35 and 30 FP). Logistic Regression misses the most (19 FN). Random Forest and XGBoost sit in between: 14 FN with 24–25 FP. In screening, a false negative may delay referral for confirmatory testing, while a false positive leads to an additional test that would likely reveal the error. The aim is therefore a low FN count without an unmanageable FP count.

4.6 ROC-AUC Comparison

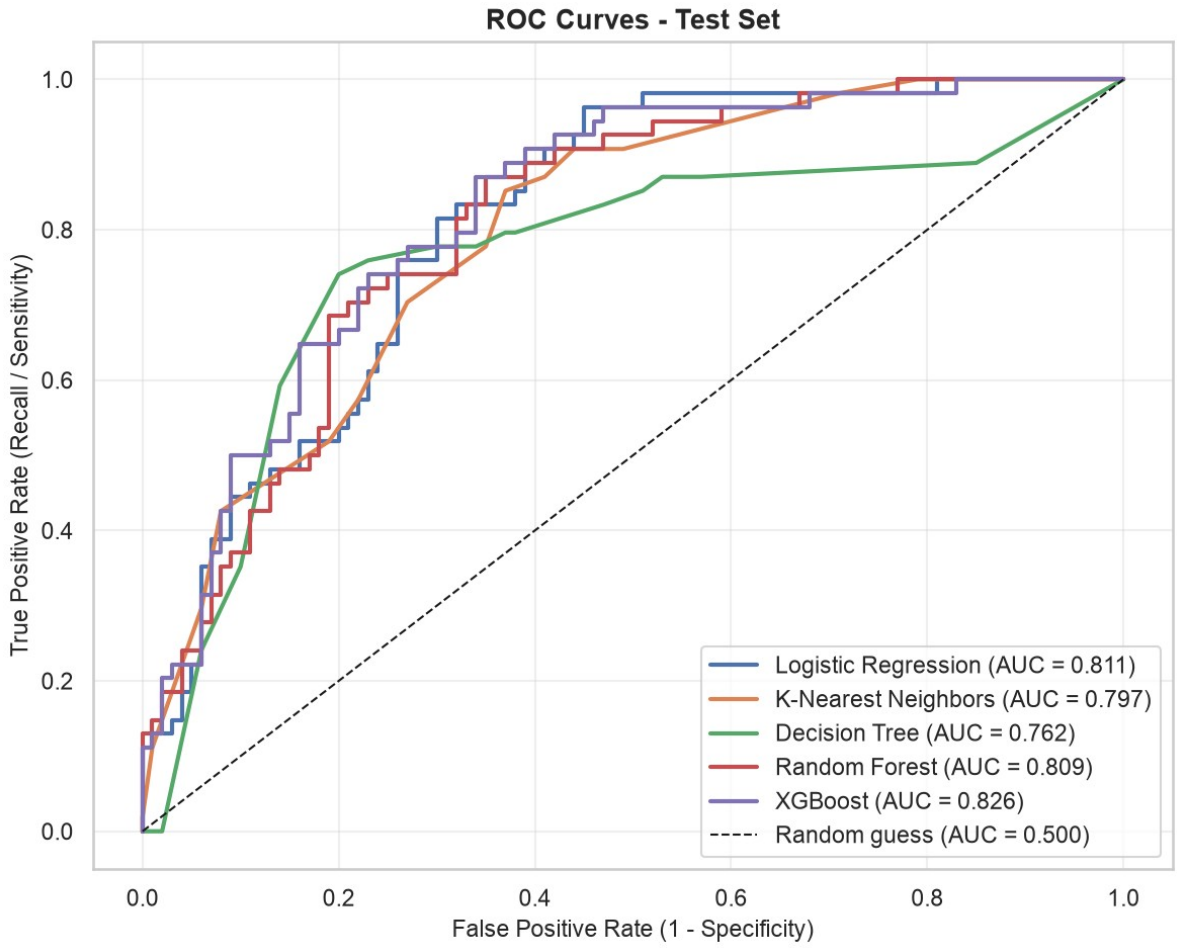


Figure 4.6: ROC curves of all models on the test set.

All models are well above the random baseline (AUC 0.5). XGBoost (0.826), Logistic Regression (0.811) and Random Forest (0.809) have nearly overlapping curves. The single Decision Tree is lowest (0.762), because a shallow tree produces only a few distinct probability values. ROC-AUC can be read as the probability that a randomly chosen diabetic patient receives a higher risk score than a randomly chosen non-diabetic patient.

4.7 Final Model Selection

Model	Test mean (5 metrics)	CV mean (5 metrics)	Test Recall	CV Recall	Overall score	Recall ≥ 0.70 in both
Random Forest	0.7169	0.7374	0.7407	0.7148	**0.7272**	**Yes**
XGBoost	0.7246	0.7096	0.7407	0.6866	0.7171	No
K-Nearest Neighbors	0.6913	0.7196	0.7778	0.7898	0.7054	Yes
Logistic Regression	0.6699	0.7372	0.6481	0.7244	0.7036	No
Decision Tree	0.7033	0.6637	0.7778	0.6870	0.6835	No

Random Forest was selected. It is the only model that ranks first or second on most metrics in *both* cross-validation and the test set, and it meets the recall constraint in both. XGBoost was best on four test metrics, but its cross-validated recall (0.687) was below the threshold. Logistic Regression had the best cross-validated ROC-AUC but the lowest test recall. KNN met the recall constraint but had clearly lower precision (0.545) and specificity (0.650).

Final model performance (test set): Accuracy 0.7468 · Precision 0.6154 · Recall 0.7407 · F1 0.6723 · ROC-AUC 0.8094 · Specificity 0.750. Cross-validated ROC-AUC: 0.829 ± 0.025.

4.8 Feature Importance

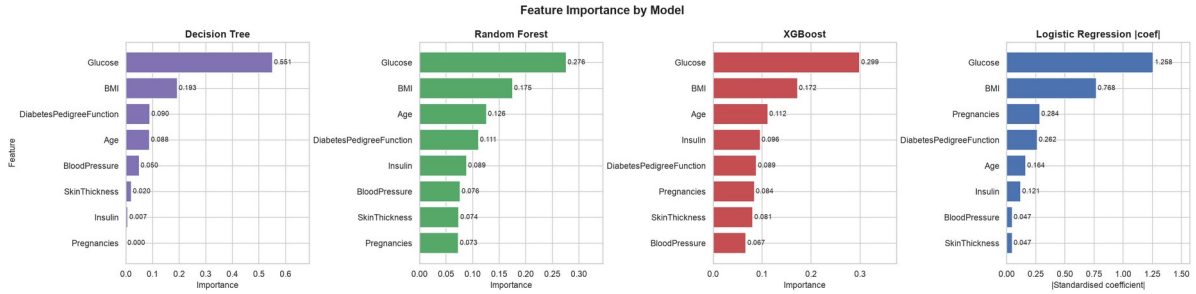


Figure 4.7: Feature importance of the tree models and |standardised coefficients| of Logistic Regression.

Feature	Decision Tree	Random Forest	XGBoost	LR |coef|
Glucose	0.551	0.276	0.299	1.258
BMI	0.193	0.175	0.172	0.768
Age	0.088	0.126	0.112	0.164
DiabetesPedigreeFunction	0.090	0.111	0.089	0.262
Insulin	0.007	0.089	0.096	0.121
BloodPressure	0.050	0.076	0.067	0.047
SkinThickness	0.020	0.074	0.081	0.047
Pregnancies	0.000	0.073	0.084	0.284

All models agree that **Glucose** is the most important feature, and **BMI** is consistently second. Age and DiabetesPedigreeFunction follow, while BloodPressure, SkinThickness and Pregnancies contribute least in the ensemble models. Impurity-based importance can favour continuous features and is shared between correlated features, so it shows how much the model *relies on* a feature, not causation.

4.9 SHAP Interpretation

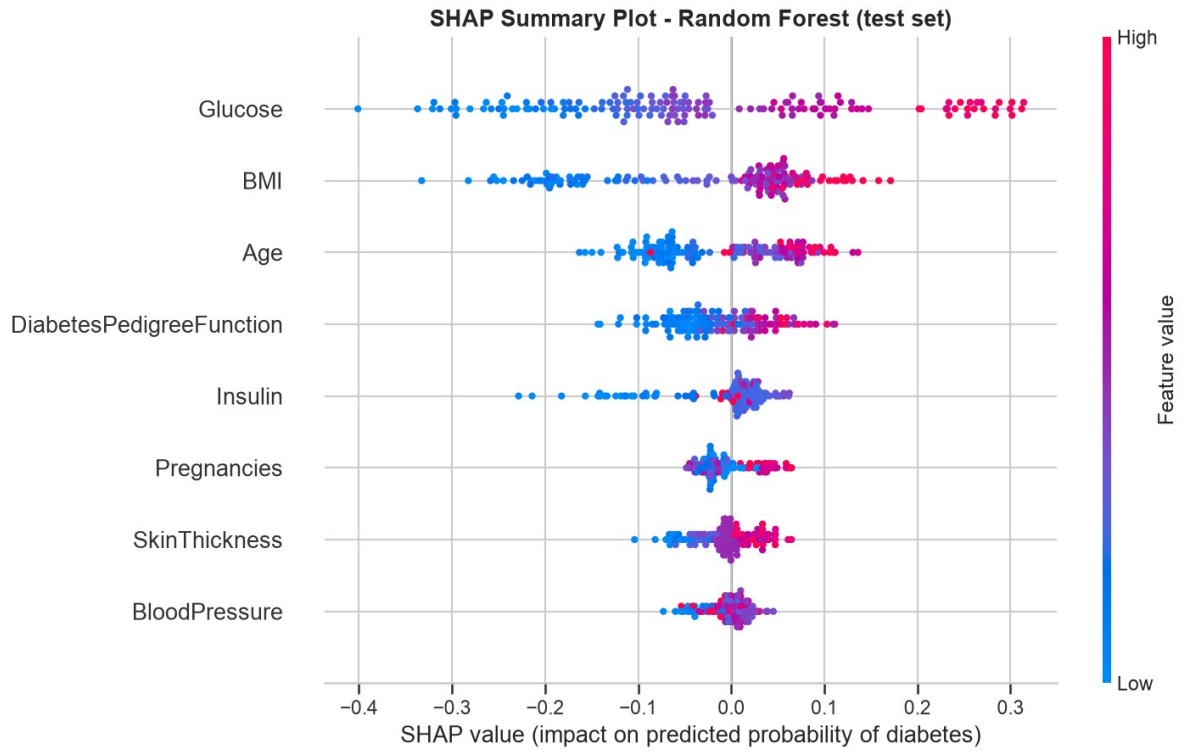


Figure 4.8: SHAP summary (beeswarm) plot for the Random Forest on the test set. Colour = feature value (red high, blue low).

Feature	Mean |SHAP|	Corr(feature value, SHAP)	Higher value pushes prediction
Glucose	0.146	0.955	toward Diabetes
BMI	0.094	0.837	toward Diabetes
Age	0.065	0.733	toward Diabetes
DiabetesPedigreeFunction	0.043	0.682	toward Diabetes
Insulin	0.034	0.237	toward Diabetes (weak)
Pregnancies	0.025	0.635	toward Diabetes
SkinThickness	0.024	0.778	toward Diabetes
BloodPressure	0.015	0.255	weak / mixed

- **Positive contributions.** High glucose has the largest positive SHAP values, raising the predicted probability by up to about 0.3. High BMI and older age also push predictions toward diabetes.
- **Negative contributions.** Low glucose, low BMI and younger age push predictions toward no diabetes. Low Insulin values produce clearly negative contributions, while high Insulin values have little effect.
- **Relative importance.** The SHAP ranking (Glucose > BMI > Age > DPF > others) matches the impurity-based ranking, and it is consistent with the EDA correlations and with the predictors reported by Joshi and Dhakal [10].

- SHAP explains how *this model* uses the features. It does not establish medical causality, and correlated features can share credit.

4.10 Sample Prediction

The `predict_diabetes()` function applies the same zero-to-NaN rule and the full fitted pipeline to one patient. Three illustrative (hypothetical) inputs gave:

Example	Key inputs	Predicted class	P(diabetes)
A	Glucose 168, BMI 38.2, Age 50	Diabetes likely	87.2 %
B	Glucose 89, BMI 24.5, Age 24	Diabetes unlikely	1.7 %
C	Glucose 130, BMI 32.0, Age 35, Insulin and SkinThickness not measured	Diabetes likely	66.7 %

These are academic demonstrations, not medical diagnoses. The probabilities are not calibrated.

4.11 Discussion

- **Glucose dominates.** Every analysis (EDA, correlation, coefficients, impurity importance and SHAP) identifies plasma glucose as the strongest signal. This agrees with the clinical role of glucose testing and with prior studies [8], [10].
- **Model families perform similarly.** Test ROC-AUCs of 0.81–0.83 for Logistic Regression, Random Forest and XGBoost suggest that the signal is largely monotonic and can be captured by a simple model. The ensembles gave better recall/precision trade-offs at the default threshold.
- **Imbalance handling matters for recall.** SMOTE raised Logistic Regression's cross-validated recall from 0.575 to 0.724 with little change in ROC-AUC.
- **CV and test can disagree.** Logistic Regression ranked first in cross-validated ROC-AUC but third on the test set, while XGBoost showed the opposite pattern. With 154 test patients, one patient changes recall by about 1.9 points, so selection based on both evaluations is more reliable than relying on the test split alone.
- **Comparison with literature.** The final model's accuracy (0.747) and AUC (0.809) are in the range of simple baselines reported on this dataset (e.g. about 76 % in [2], [7], and a random-forest accuracy of 0.7604 on complete Pima records in [8]). They are below heavily engineered pipelines [9], which use additional steps (outlier removal, feature selection, ensembling) and different validation schemes. Because protocols differ, these numbers are not directly comparable.

Chapter 5 — Conclusion

5.1 Summary

This project built a complete, leak-free predictive analytics pipeline for diabetes-onset prediction on the Pima Indians Diabetes Dataset, following the KDD process from problem definition to knowledge discovery. Invalid zeros were treated as missing, the data were split with stratification before any fitting, and imputation, scaling and SMOTE were learned from the training data only. Five classifiers were compared with cross-validation and a held-out test set.

5.2 Main Findings

- All five models were far better than random guessing (test ROC-AUC 0.762–0.826).
- Handling class imbalance substantially improved recall at a small cost in accuracy.
- Simple and complex models performed similarly in ranking ability. The ensembles gave the best balance of recall and precision at the default threshold.

5.3 Best Model

Random Forest (median imputation → scaling → SMOTE → 300 trees) was selected with a recall-aware, multi-metric rule. Test results: **accuracy 0.747, precision 0.615, recall 0.741, F1 0.672, ROC-AUC 0.809**. It correctly flagged 40 of 54 diabetic patients, with cross-validated ROC-AUC 0.829 ± 0.025.

5.4 Important Features

Glucose is the dominant predictor, followed by **BMI**, **Age** and the **Diabetes Pedigree Function**. Higher values of these features are associated with higher predicted risk. These are predictive associations, not causal effects.

5.5 Limitations

- Small dataset (768 patients; 154 in the test set), so metrics carry substantial uncertainty.
- Narrow demographics (Pima Indian women aged 21 or older) and historical data; generalisation to other populations is unknown.
- Possible sampling bias from a single population study.
- Heavy missingness in Insulin (48.7 %) and SkinThickness (29.6 %), filled by median imputation.
- SMOTE samples are synthetic, not real patients.
- No exhaustive tuning, a fixed 0.5 threshold, uncalibrated probabilities and no external validation.
- The model is an academic screening demonstration, not a medical diagnostic tool.

5.6 Future Scope

Larger and more diverse datasets; external validation; cross-validated or nested hyperparameter optimisation; per-patient explanations (SHAP waterfall plots); probability calibration and cost-sensitive threshold selection; additional clinical features such as HbA1c; deployment as a screening-*support* tool with appropriate oversight; and monitoring for data and model drift after deployment.

5.7 KDD Mapping

KDD Stage	Implementation in Project
Problem Definition	Diabetes onset prediction (binary classification)
Data Selection	Pima Indians Diabetes Dataset
Data Cleaning	Invalid zero handling (5 features → NaN)
Data Transformation	Stratified split, median imputation, scaling, SMOTE (train only)
EDA	Distributions, by-outcome plots, boxplots, correlations, pairplot
Modeling	LR, KNN, DT, RF, XGBoost
Evaluation	Accuracy, Precision, Recall, F1, ROC-AUC, confusion matrices, CV
Interpretation	Feature importance + SHAP
Knowledge Discovery	Glucose, BMI and Age as key predictive patterns; Random Forest as a balanced screening model

Chapter 6 — References

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